Thesis Presentation:

Forecasting with Linear Augmented Information Combination

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EBS Honours/Masters Presentation

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Outline

Part 1. *Research Question and Method*

1.1 Forecast reconciliation

1.2 Multivariate regression framework with flexible predictor set

Forecasting with Linear Augmented Information Combination

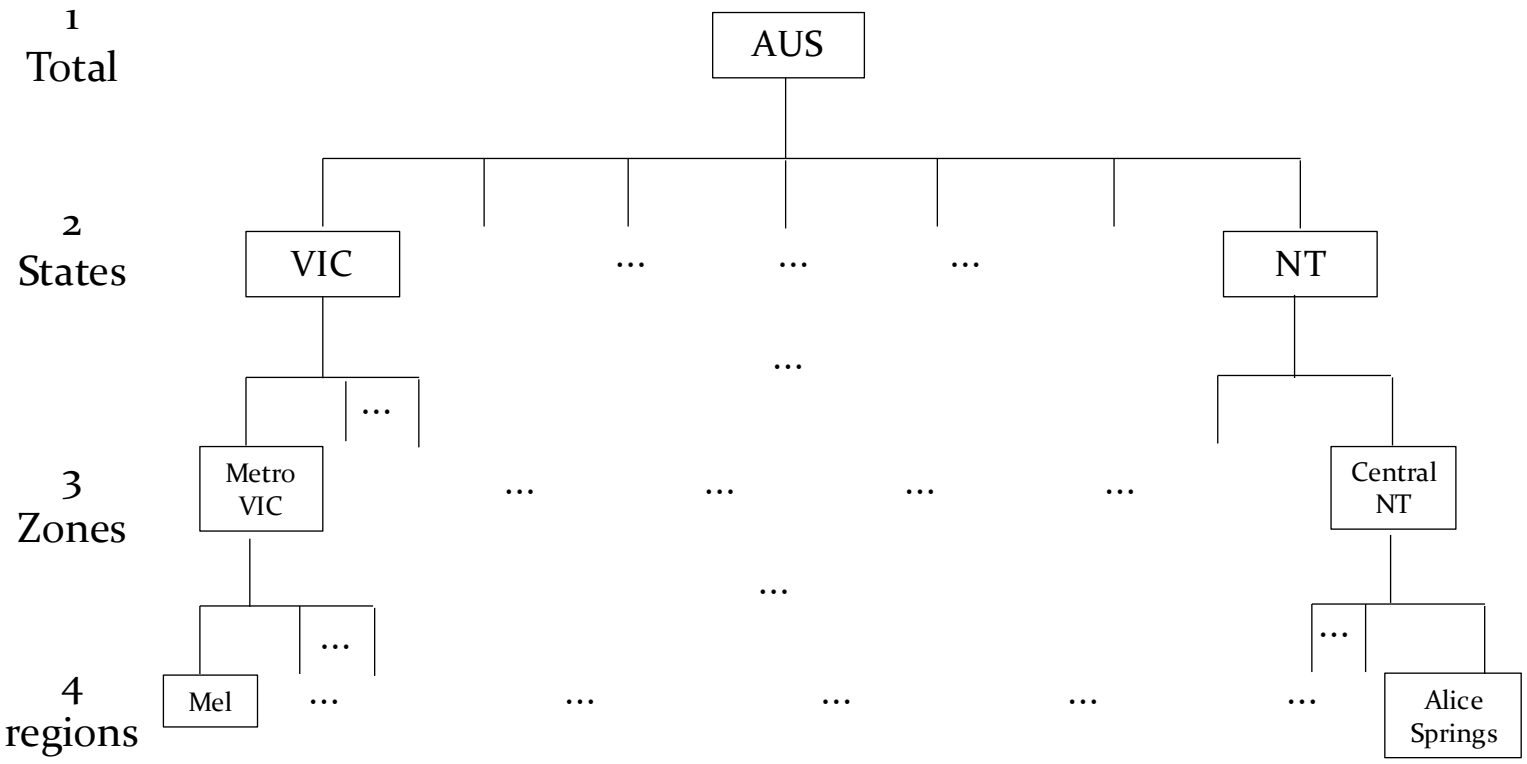
1.3 Linear components (FLAP idea) augmented in predictor set

Part 2. *Empirical Investigation*

1.1 What is forecast reconciliation problem?

Levels

Domestic tourism hierarchy *(n=112 series)*



Hierarchical structure of data

$$\mathbf{y}_t = \mathbf{S}\mathbf{b}_t$$

Aggregation Constraints

$$\begin{bmatrix}\mathbf{a}_t\\ \mathbf{b}_t\end{bmatrix} = \begin{bmatrix}\mathbf{A}\\ \mathbf{I}_{n_b}\end{bmatrix}\mathbf{b}_t$$

$$\mathbf{C}\,\mathbf{y}_t = [\mathbf{I}_{n_a} \;\; -\mathbf{A}]\,\mathbf{y}_t = \mathbf{0}$$

Forecasting problem:

Base forecasts (h-step ahead) using

$$\mathfrak{T}_T = (\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_T)$$

are *incoherent*

$$\hat{\mathbf{y}}_{T+h|t} \neq \mathbf{S}\,\hat{\mathbf{b}}_{T+h|t}$$

Base forecasts $\hat{\mathbf{y}}_{T+h|t}$ ⟶ Reconciliation ⟶ Coherent $\tilde{\mathbf{y}}_{T+h|t}$

1.2 How has the literature approached reconciliation?

Unbiasedness of base forecasts assumption:

$$\mathrm{E}\left[\hat{\mathbf{y}}_{t+h|t}|\mathfrak{I}_t\right] = \mathrm{E}[\mathbf{y}_{t+h}|\mathfrak{I}_t]$$

Strong requirement (Nguyen et al. 2025)

$$\Rightarrow \quad \mathrm{E}\left[\mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t}\,|\mathfrak{I}_t\right] = \mathrm{E}\left[\hat{\mathbf{e}}_{t+h|t}\,|\mathfrak{I}_t\right] = 0$$

$$\Rightarrow \quad \hat{\mathbf{y}}_{t+h|t} = \mathrm{E}[\mathbf{y}_{t+h}|\mathfrak{I}_t] + \boldsymbol{\epsilon}, \quad \mathrm{E}[\boldsymbol{\epsilon}|\mathfrak{I}_t] = 0$$

Drawback: assumes base forecasts embody ***all*** useful information in history of the hierarchy

Minimum Trace (MinT) (Wickramasuriya et al., 2019)

- Unbiasedness assumption + Optimisation
- Find optimal $\mathbf{G}_h$ for:

$$\tilde{\mathbf{y}}_{T+h|T} = \mathbf{S}\,\tilde{\mathbf{b}}_{T+h|T} = \mathbf{S}\,\mathbf{G}_h\,\hat{\mathbf{y}}_{T+h|T}$$

- Solution is: $\mathbf{G}_{h,\mathrm{MinT}} = \left(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{W}_h^{-1}$

where: $\mathbf{W}_h = \mathrm{Var}(\hat{\mathbf{e}}_{t+h|t}), \quad \hat{\mathbf{e}}_{t+h|t} = \mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t}$

Information Combination (IComb) (Nguyen et al. 2025)

- Multivariate regression: $\mathbf{Y} = \mathbf{X}\,\mathbf{B} + \mathbf{E}$

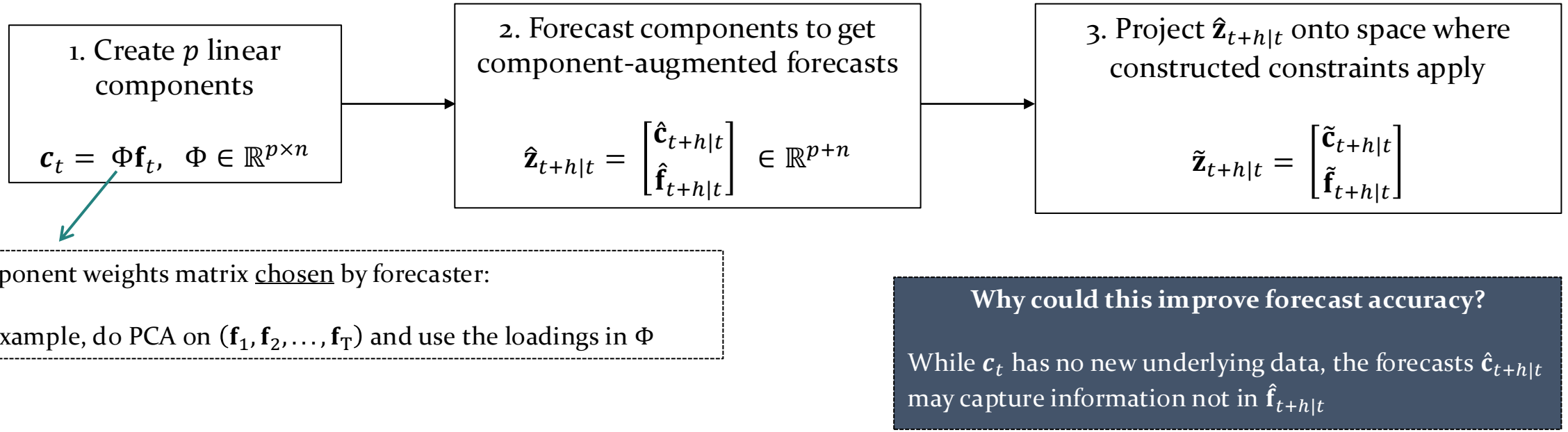
  where: $\mathbf{Y}$: coherent realised hierarchy

  $\mathbf{X}$: *any* predictor set, hierarchy in-sample forecasts

- $\tilde{\mathbf{y}}_{T+h|T} = \hat{\mathbf{B}}'\,\hat{\mathbf{y}}_{T+h|T}$ coherent as: $\mathbf{C}\,\hat{\mathbf{B}}' = \mathbf{C}\,\mathbf{Y}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \mathbf{0}$
- Regularisation:
  - IComb-S: Ridge (shrinkage) → dense models
  - IComb-L: Multivariate lasso → sparse models

1.3 Gap: how to expand the predictor set beyond forecasts of $\mathbf{y}_t$?

- Forecast Linear Augmented Projection (FLAP) (Yang et al., 2024) says you can improve forecast accuracy of a multivariate series $\mathbf{f}_t \in \mathbb{R}^n$ by creating linear components from it and forecasting them too



Research Question: *To what extent* can forecasts based on constructed linear components improve the accuracy of hierarchical forecasts when incorporated into the Information Combination framework?

Forecasting with Linear Augmented Information Combination

1. Create p linear component series from $\mathfrak{T}_T$ using PCA and forecast them → 2. Apply IComb framework on set of augmented predictors, to get FLAIC coefficient matrix → 3. Use obtained coefficients to reconcile base forecast

- We use all *four* PCA specifications. Let $\delta_C, \delta_S \in \{0,1\}$ binary indicators for centring and scaling
- Define

$$\mathbf{y}_t^{(\delta_C,\delta_S)} = \mathbf{D}_{(\delta_C)}^{-\delta_S}(\mathbf{y}_t - \delta_C\bar{\mathbf{y}})$$

  where: $\mathbf{D}_{(\delta_C)}$ is diagonal matrix of scaling factors

- Apply PCA to $\mathbf{Y}_{\mathrm{T}}^{(\delta_C,\delta_S)}$ construct $p \leq \min(T-h-\delta_C, n_b)$ components as:

$$\mathbf{c}_t^{(\delta_C,\delta_S)} = \mathbf{\Phi}_p^{(\delta_C,\delta_S)}\left(\mathbf{D}_{(\delta_C)}^{-\delta_S}(\mathbf{y}_t - \delta_C\bar{\mathbf{y}})\right)$$

PC scores *PC Loadings*

- Forecast components, $\hat{\mathbf{c}}_{t+h|t}$

Forecasting with Linear Augmented Information Combination ...*(cont'd)*

- Create an augmented predictor set, $\mathbf{Z}$, for responses, $\mathbf{Y}$

$$\hat{\mathbf{z}}_{t+h|t} = \begin{bmatrix} \hat{\mathbf{c}}_{t+h|t} \\ \hat{\mathbf{y}}_{t+h|t} \end{bmatrix} \qquad \text{stacked:} \qquad \mathbf{Z} = \begin{bmatrix} \hat{\mathbf{z}}'_{1+h|1} \\ \hat{\mathbf{z}}'_{2+h|2} \\ \vdots \\ \mathbf{z}'_{T|T-h} \end{bmatrix}, \quad \text{with} \quad \mathbf{Y} = \begin{bmatrix} \mathbf{y}'_{1+h} \\ \mathbf{y}'_{2+h} \\ \vdots \\ \mathbf{y}'_{T} \end{bmatrix}$$

- Apply IComb framework, replacing $\mathbf{X}$ with $\mathbf{Z}$ only: $\mathbf{Y} = \mathbf{Z}\,\mathbf{B} + \mathbf{E}$
- Reconciled $\tilde{\mathbf{y}}^{FLAIC}_{\mathrm{T}+h|\mathrm{T}} = \left(\widehat{\mathbf{B}}^{\mathrm{FLAIC}}\right)' \hat{\mathbf{z}}_{\mathrm{T}+h|\mathrm{T}}$ is coherent

- Challenges motivate regularisation
  1. Additional p coefficients to be estimated in each of the n regressions
  2. Multicollinearity: augmented predictor set may have highly correlated or redundant signals
     - ✓ FLAIC-S: Ridge (shrinkage) → dense models
     - ✓ FLAIC-L: Multivariate lasso → sparse models

- Potential improvement in forecast accuracy:
  1. The forecasts, $\hat{\mathbf{c}}_{t+h|t}$ may capture information not in $\hat{\mathbf{y}}_{t+h|t}$
  2. If $\mathbf{c}_t$ exhibits more forecastable dynamics, $\hat{\mathbf{c}}_{t+h|t}$ may provide complementary information

2.0 Empirical Investigation

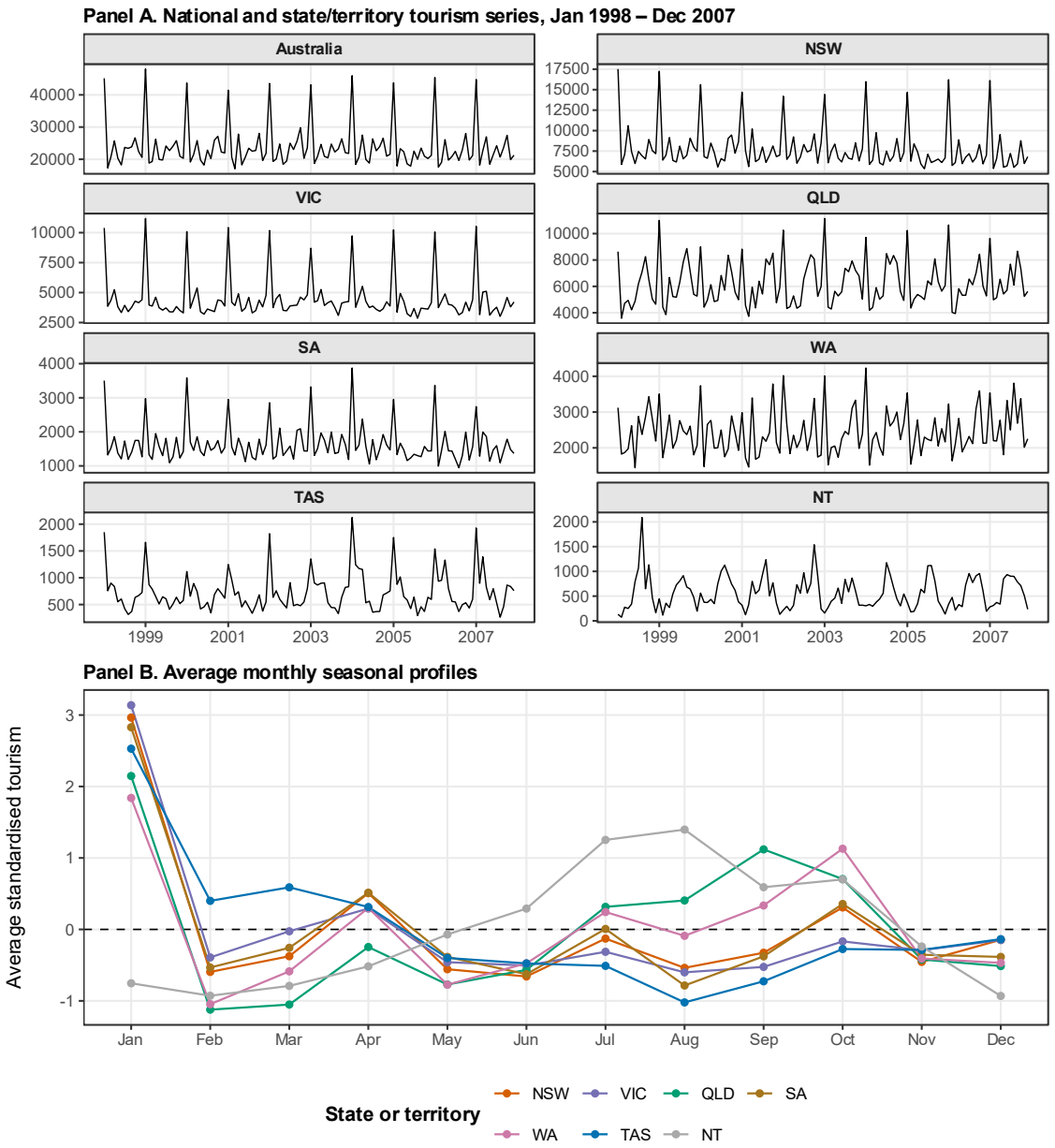


Figure. *Australian domestic tourism series and seasonal profiles*

- Australian domestic tourism visitor nights (Tourism Research Australia, 2026)
- 4 levels: national, states & territories, zones, regions
- 112 series, $n_b = 77$
- 264 months: Jan 1998 – Dec 2019
- Forecasting this data (and components):
  - 120 month rolling windows
  - Error-Trend-Seasonality (ETS) models
  - $h = 1,2, \dots, 12$ month ahead out-of-sample (short-term: $h = 1 - 6$ & long-term $h = 1 - 12$)
  - $144 - h$ available forecast originsx

Components created for all four PCA (centre, scale) specifications

PCA specification	PC	Share (%)	Seasonal R^2 (%)
Centred, unscaled	PC1	93.1 [92.6, 93.5]	93.8 [74.7, 96.0]
	PC2	3.4 [3.1, 3.7]	79.7 [77.2, 83.2]
	PC3	0.8 [0.5, 1.4]	24.9 [8.6, 34.0]
Centred, scaled	PC1	38.7 [38.0, 40.0]	94.4 [78.2, 95.8]
	PC2	13.3 [12.2, 14.9]	86.0 [72.9, 89.4]
	PC3	3.2 [2.8, 3.9]	39.4 [15.3, 58.1]
Uncentred, unscaled	PC1	99.3 [99.3, 99.4]	93.5 [74.0, 95.8]
	PC2	0.4 [0.3, 0.4]	91.1 [89.1, 93.3]
	PC3	0.1 [0.1, 0.1]	42.9 [6.7, 52.3]
Uncentred, scaled	PC1	81.6 [81.2, 83.8]	90.2 [69.3, 92.6]
	PC2	5.6 [5.1, 5.9]	94.4 [92.9, 94.9]
	PC3	1.0 [0.9, 1.1]	68.0 [64.0, 71.0]

Table: Median [2.5th, 97.5th percentiles] across rolling origins. Share denotes percentage of total sum of squares. Seasonal R^2 is from month-of-year regressions.

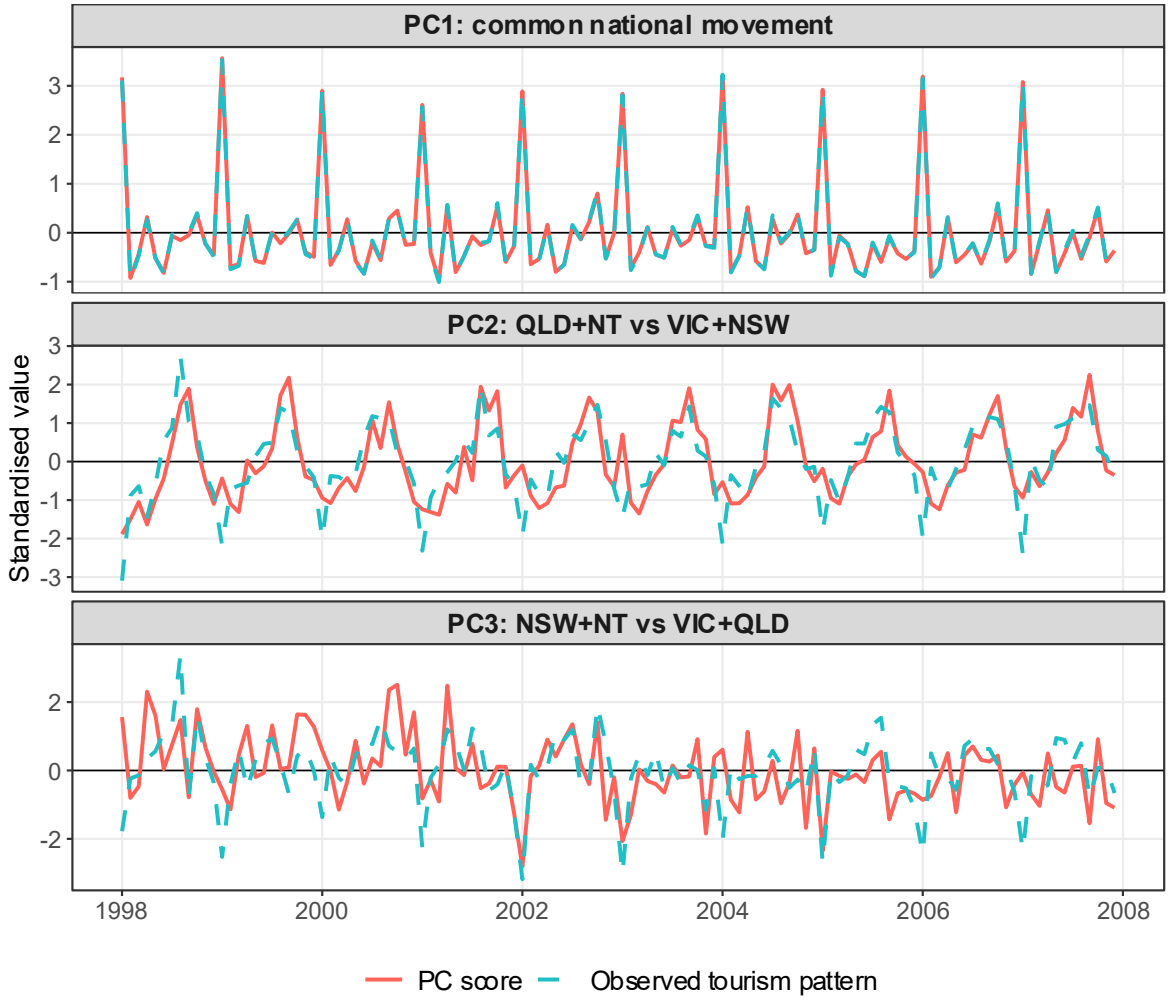


Figure: Illustrative patterns from centred, unscaled PCA using the initial training window. Series are standardised for visual comparison.

Each rolling training window:

- Conventional method reconciliation: BU, OLS, WLS_v, WLS_s, MinT_{shr}

- IComb
  - Ridge (IComb-S) and lasso (IComb-L) regularisation approaches
  - IComb variant
    - $\mathbf{X}$ and $\mathbf{Y}$ are standardized: $\mathbf{X}^* = \mathbf{X}\,\boldsymbol{\Sigma}_x^{-0.5}$, $\mathbf{Y}^* = \mathbf{Y}\,\boldsymbol{\Sigma}_y^{-0.5}$, $\hat{\mathbf{B}} = \boldsymbol{\Sigma}_x^{-0.5}\,\hat{\mathbf{B}}^*\,\boldsymbol{\Sigma}_y^{0.5}$
  - Tuning parameter (λ) selected based on expanding window cross-validation
    - Grid size = 200
    - Inner window's validation set size = 40
    - Optimal λ minimises mean squared forecast error of validation set

- FLAIC
  - 64 variants:

$$4 \text{ PCA specifications} \quad \times \quad p \in \{\underbrace{1,2,3,5}_{\text{low}},\ \underbrace{10,20}_{\text{moderate}},\ \underbrace{40,77}_{\text{high}}\} \quad \times \quad \{\text{FLAIC-S, FLAIC-L}\}$$

  - Component augmented predictor set in IComb with:
    - $\mathbf{Z}$ and $\mathbf{Y}$ are standardized; intercept included and not regularized
    - Tuning parameter (λ) selected based on expanding window cross-validation

2.4.1 PRIAL

- Forecast performance measure: percentage relative improvement in average loss (PRIAL) (Nguyen et al., 2025):

- Let m index method. MSFE computed over all series (Average) and rolling forecast origins.

$$\text{PRIAL}_{m,h} = \frac{\text{MSFE}_{m,h} - \text{MSFE}_{\text{base},h}}{\text{MSFE}_{\text{base},h}} \times 100$$

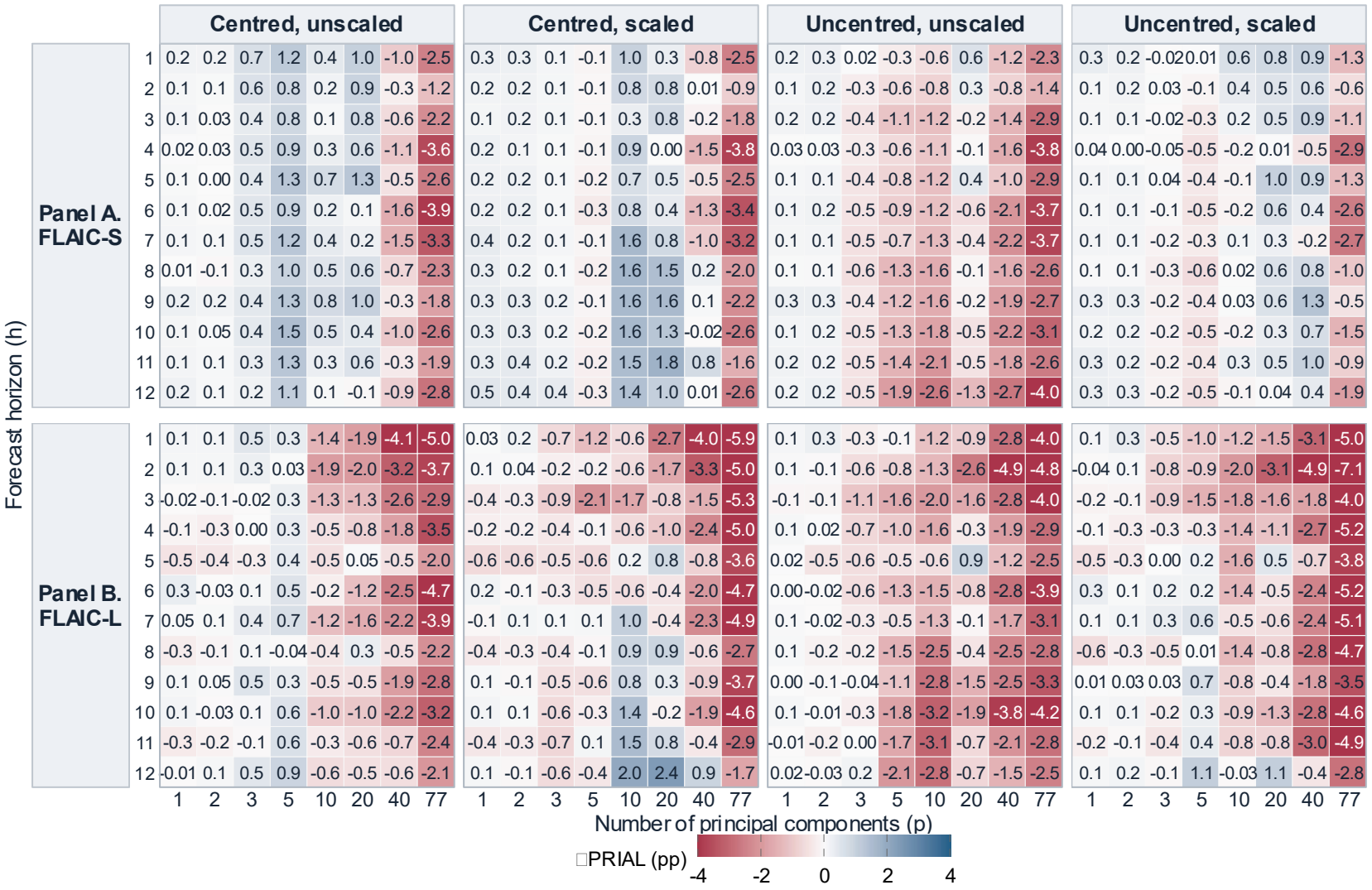
- Key findings:
  - Best *overall*: FLAIC-L
    - Earlier horizons ($h = 1$ to 6): centred, unscaled ($p = 3$ or 5)
    - Later horizons ($h = 7$ to 12): centred, scaled ($p = 10$ or 20)
  - Best FLAIC-S has similar PCA specification and p as FLAIC-L
  - Does not necessarily indicate component augmentation more effective under lasso

PRIAL for long-term period ($h = 1 - 12$):

Regularisation/Method	IComb	FLAIC	Difference
Ridge	-4.65%	-5.82%	1.17 pp
Lasso	-6.25%	-6.67%	0.42 pp
Difference	1.6 pp	0.85 pp	

* *Best FLAIC-L:* centred, unscaled, p=5
* *Best FLAIC-S:* centred, scaled, p=10

2.4.2 Δ PRIAL



- When does component augmentation improve upon IComb?

$$\Delta\text{PRIAL}_h = \text{PRIAL}_{\text{IComb},h} - \text{PRIAL}_{\text{FLAIC},h}$$

$$= \frac{\text{MSFE}_{\text{IComb},h} - \text{MSFE}_{\text{FLAIC},h}}{\text{MSFE}_{\text{base},h}} \times 100$$

- $\Delta\text{PRIAL} > 0 \implies$ FLAIC better
- Findings:
  1. Effects of component augmentation depends on p
  2. Component augmentation generally more effective for ridge than lasso
  3. Direction of relative performance often maintained across h

Figure: *ΔPRIAL*: FLAIC relative to matched IComb: Average across 112 hierarchy series. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L.

2.4.2 Δ PRIAL *(decomposed)*

- PRIAL depends on MSFE.

$$\text{MSFE}_{m,h} = \frac{1}{N_h n} \sum_{\tau \in \mathcal{O}_h} \sum_{i=1}^{n} \left(\tilde{e}_{\tau,h,i}^{(m)} \right)^2 = \frac{1}{N_h} \sum_{\tau \in \mathcal{O}_h} \ell_{\tau,h}^{(m)}$$

- Therefore, *after averaging across series*:

$$\Delta\text{PRIAL}_h = \frac{\text{MSFE}_{\text{IComb},h} - \text{MSFE}_{\text{FLAIC},h}}{\text{MSFE}_{\text{base},h}} \times 100$$

$$= \frac{1}{N_h} \frac{\sum_{\tau \in \mathcal{O}_h} 100 \left(\ell_{\tau,h}^{(\text{IComb})} - \ell_{\tau,h}^{(\text{FLAIC})} \right)}{\text{MSFE}_{\text{base},h}}$$

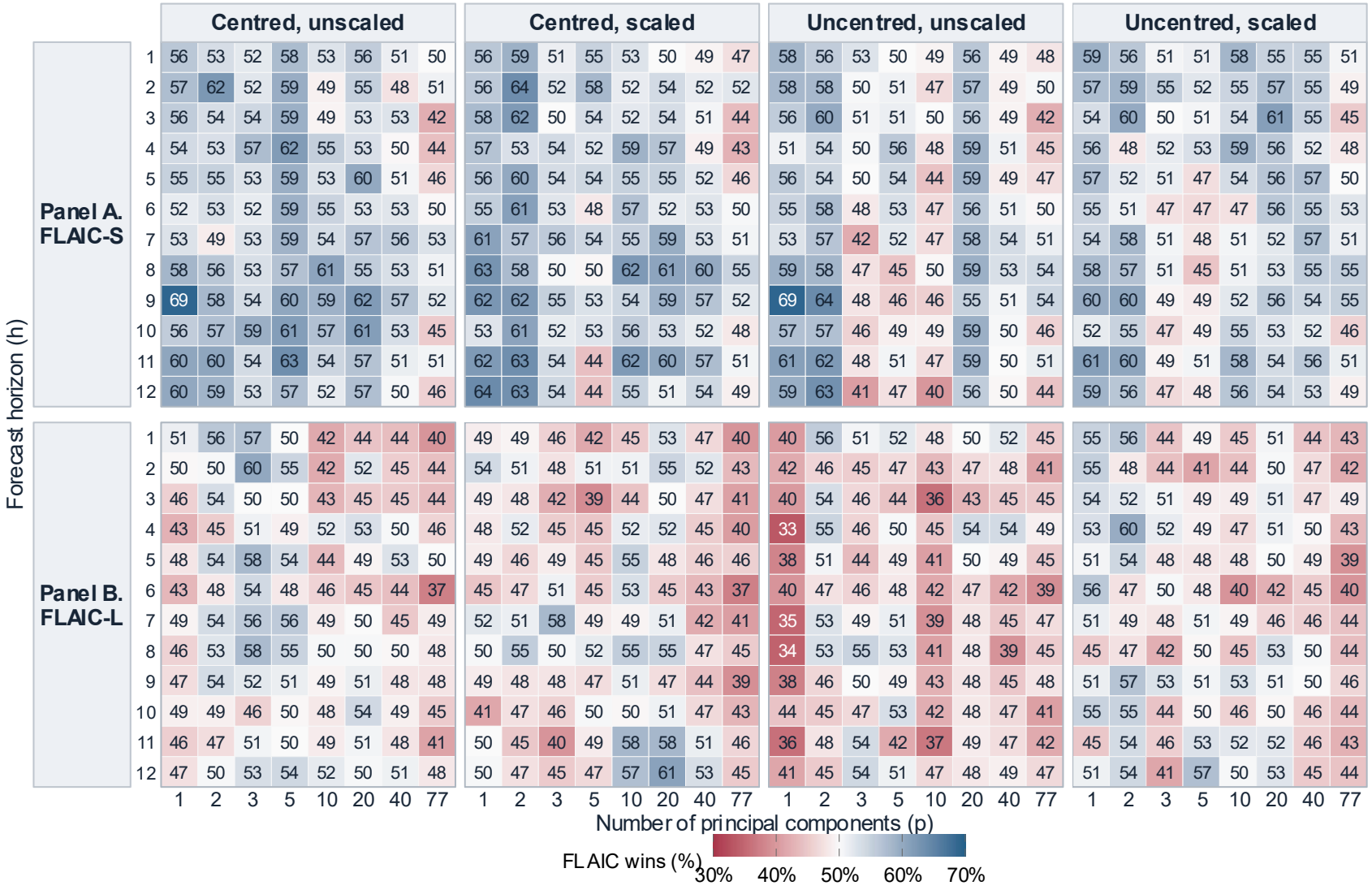
(2.4.3)
Origin-level loss difference > 0
$\Longrightarrow$ FLAIC better for that origin

Statistic: % origins FLAIC better
>50% $\Longrightarrow$ FLAIC better more origins

(2.4.4)
$q_{\tau,h}$ = origin-level contributions to ΔPRIAL_h (magnitudes across origins)

- We can also first average squared errors across rolling window origins, then compare for series
  - Natural series grouping by levels *(2.4.5)*

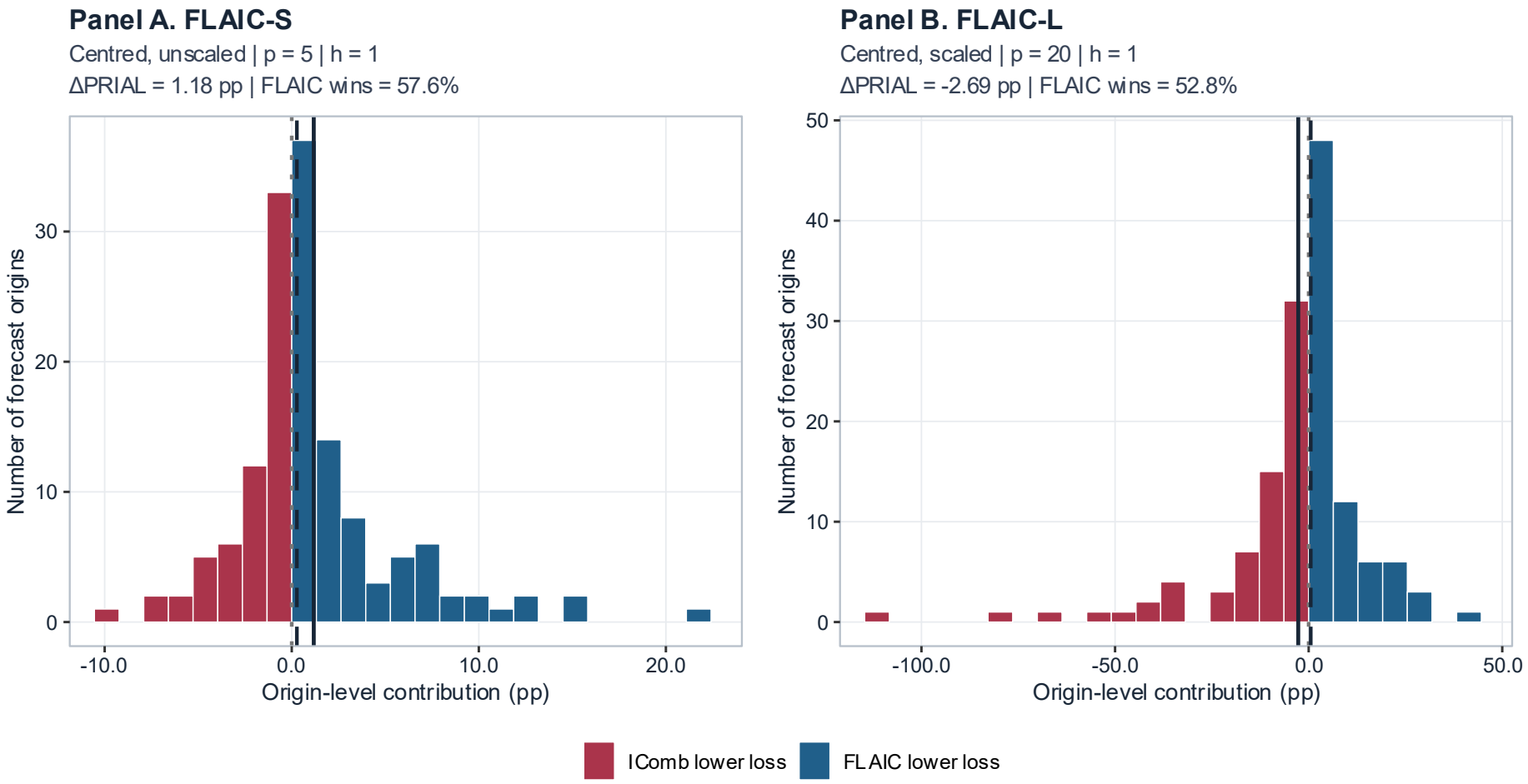
2.4.3 Rolling forecast origin wins (FLAIC v IComb)



- Agrees with ΔPRIAL_h findings
- Comparison by cell for both heatmaps
  - ΔPRIAL and FLAIC wins match 74.2% (FLAIC-S) and 66.9% (FLAIC-L)
  - Most differences arise when ΔPRIAL < 0, but FLAIC wins > 50%
- We consider magnitudes by origin for two cases under $h = 1$:
  1. **A:** FLAIC-S, centred, unscaled, $p = 5$ ($\Delta\text{PRIAL} = 1.2$pp, FLAIC wins = 58%)
  2. **B:** FLAIC-L, centred, scaled, $p = 20$ ($\Delta\text{PRIAL} = -2.7$pp, FLAIC wins = 53%)

Figure: Percentage of rolling forecast origins at which FLAIC has a lower average squared-error loss than matched IComb, averaged across the 112 hierarchy series. FLAIC-S and FLAIC-L are compared with IComb-S and IComb-L, respectively. Percentages are rounded to integers.

2.4.4 Origin–level contributions to Δ PRIAL



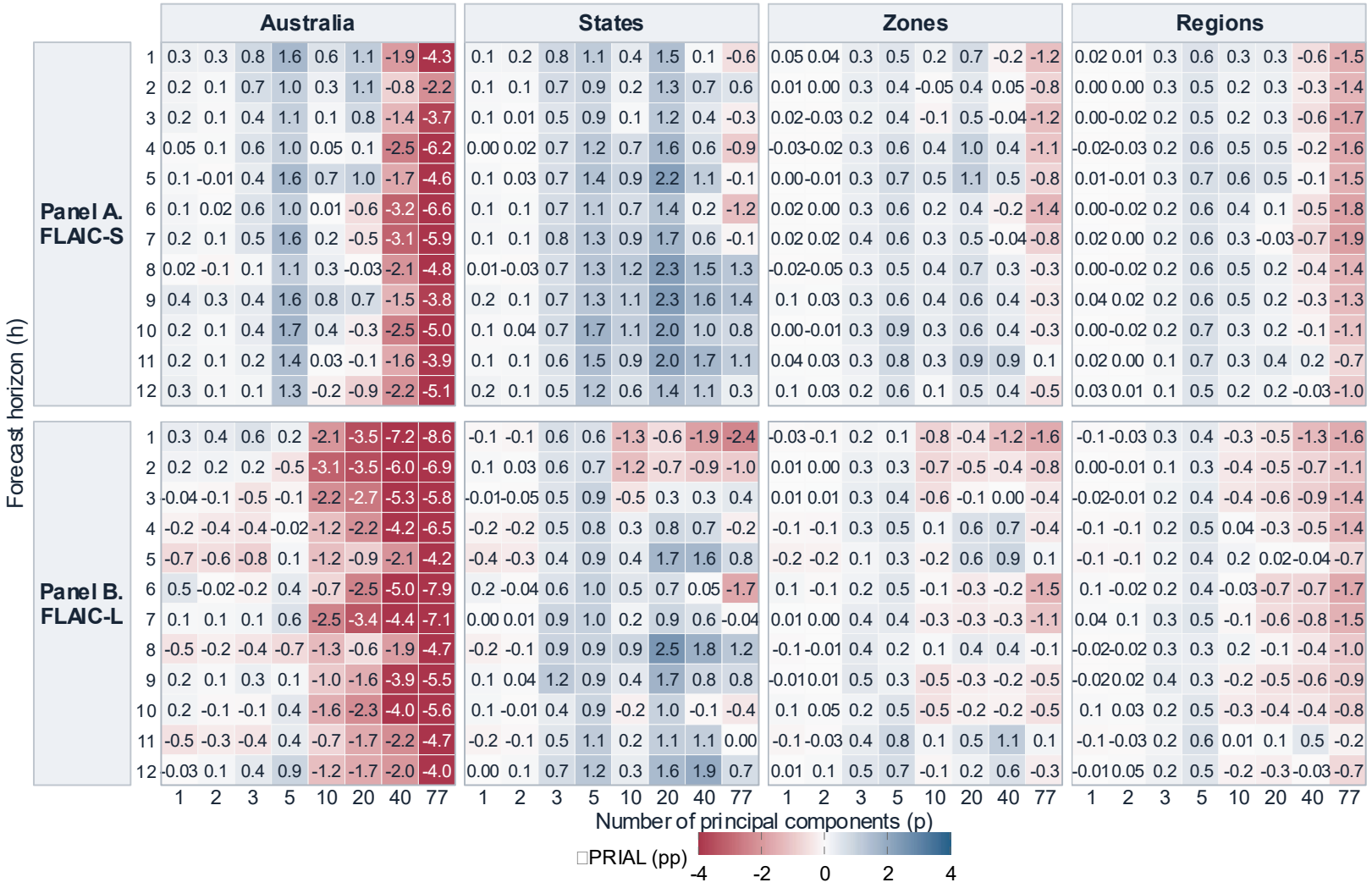
Vertical lines: solid = mean (ΔPRIAL); dashed = median; dotted = zero.

Figure: Distributions of origin-level contributions to Δ *PRIAL* at $h = 1$, averaged across the 112 hierarchy series. Positive values favour FLAIC.

Panel A. FLAIC performs better over greater number of origins and distribution of mean squared loss is relatively even

Panel B. High magnitude of severely poor performance of FLAIC relative to IComb outweighs many origins where FLAIC performs relatively better than IComb

2.4.5 Δ PRIAL and PRIAL by hierarchy level



- Heterogeneous effects across hierarchy levels
  - ✓ States tend to benefit more
- Higher dimensional component augmentation mostly deteriorates relative forecast performance
- Relative to base forecast (PRIAL):
  - Best PCA specification, p, ridge/lasso vary by level

Figure: *ΔPRIAL*: FLAIC relative to matched IComb (using centred, unscaled PCA), computed for each hierarchy level. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L, and positive values favour FLAIC over matched IComb.

2.4.6 Statistical significance visualised via MCB

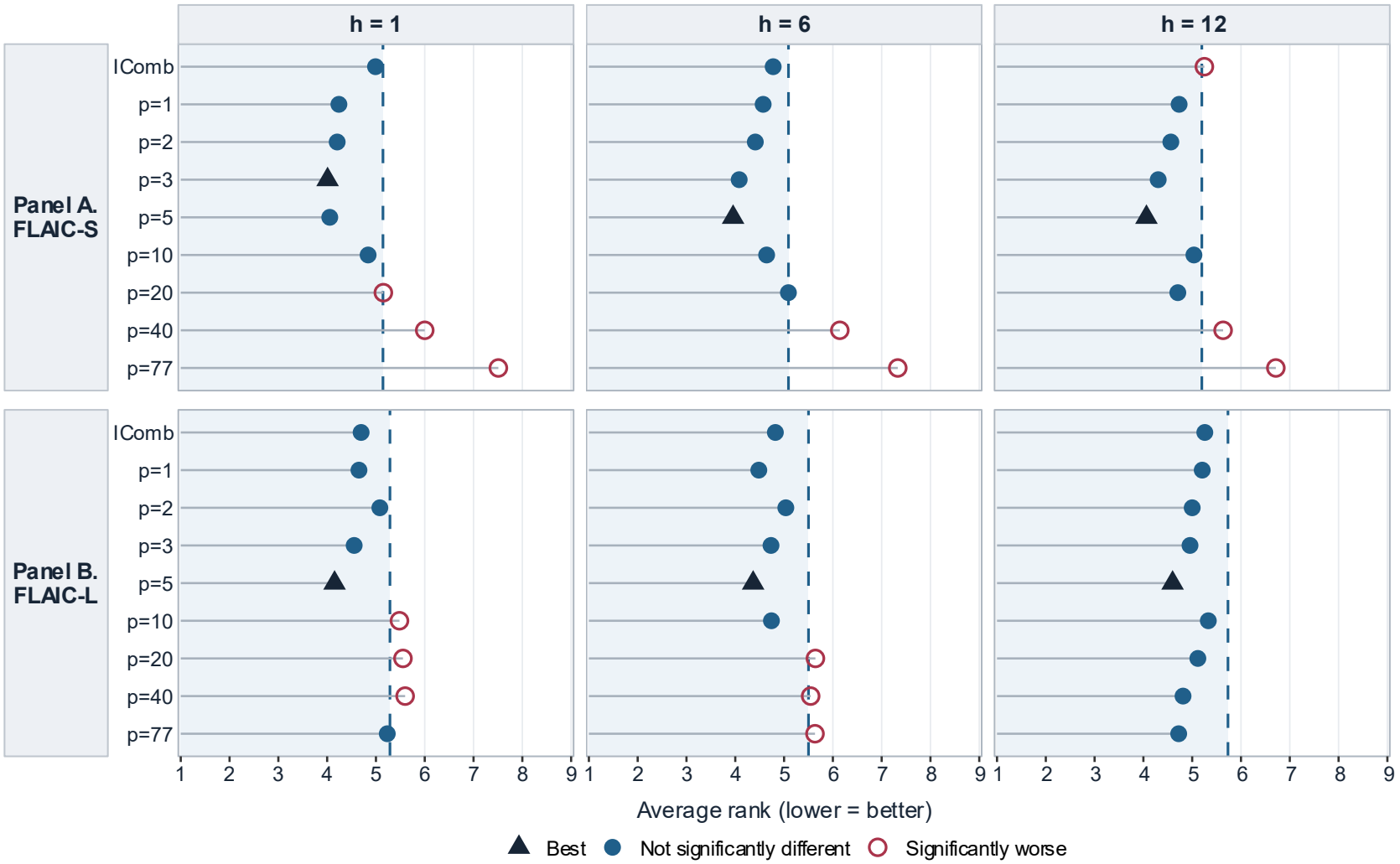


Figure: Multiple comparison with best (MCB): comparison of average ranks against best-rank method. Uses average MSFE ranks across 112 hierarchy series at $h = 1,6,12$ for centred, unscaled PCA specification.

$$\text{MSFE}_{m,h,i} = \frac{1}{N_h} \sum_{\tau \in \mathcal{O}_h} \left(\tilde{e}_{\tau,h,i}^{(m)} \right)^2$$

- Rank each series across methods to get a distribution of ranks for each method. Each method is summarised by its average rank across series.
  (Friedman 1937, 1939; Nemenyi 1963; Hollander et al. 2013; Koning et al. 2005)

Considering other 3 PCA specs too:

- Best-ranked method generally has low component augmentation
- Limited evidence that improvement in forecast accuracy under FLAIC is statistically significant (especially lasso)
- High-dimensional component augmentation is typically statistically significantly worse (especially ridge)

2.4.7 *Lasso diagnostics*

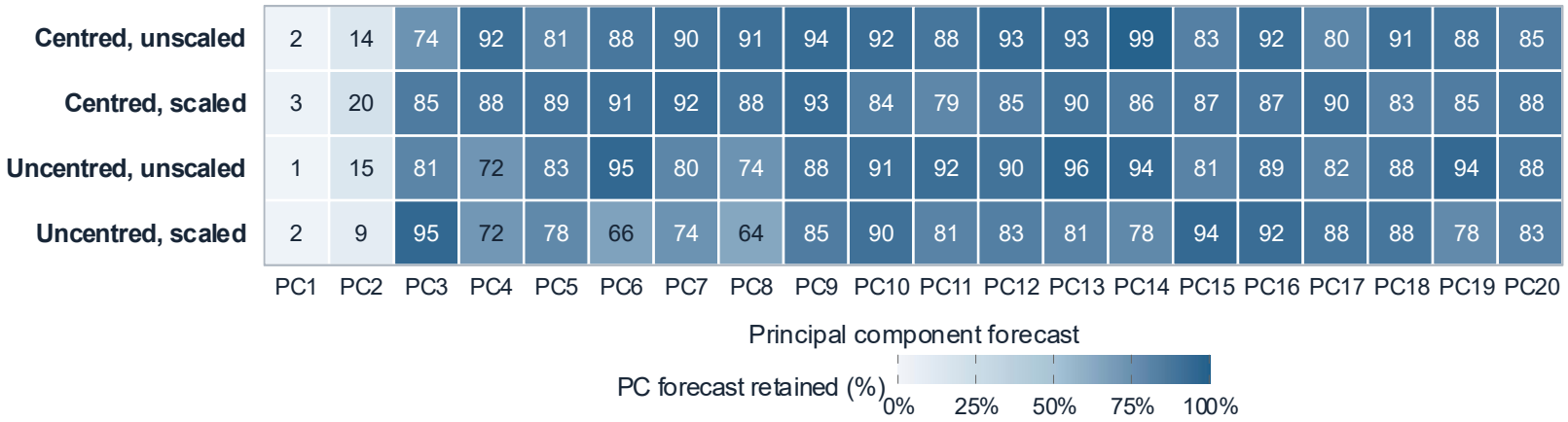


Figure: Cells report the percentage of 144 rolling forecast origins at which each PC forecast is retained by FLAIC-L. Retention means at least one non-zero coefficient across the 112 response equations.

Component forecasts retained (median) by FLAIC-L: (number, %)

16 / 20 (80%)

Δ hierarchy base forecasts retained relative to IComb-L median:

– 2

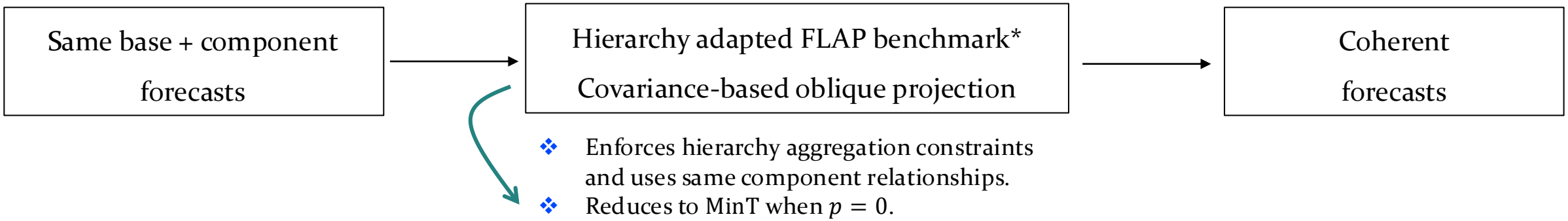
Example: At $p = 20$, centred and unscaled PCA

*Note: IComb-L retained a median of 52 of 112 hierarchy base forecasts

Findings for component forecast retention:

- PC1 and PC2 rarely retained while higher order PCs frequently kept
- Component forecast retention is not accompanied by a comparable reduction in hierarchy base forecasts.
- Retention does not consistently translate into accuracy gains.

2.4.8 Hierarchy-adapted FLAP benchmark



Extract: component augmentation across reconciliation methods

Centred, Scaled PCA | PRIAL (%) | $h = 1$

Method	**No PCs**	$p = 5$	$p = 10$	$p = 20$	$p = 40$	$p = 77$
$\text{MinT}_{shr} \rightarrow$ FLAP	5.00	1.16	0.35	-0.29	-0.55	-0.08
IComb − S → FLAIC − S	−4.98	−4.85	−5.97	−5.26	−4.15	−2.45
IComb − L → FLAIC − L	−7.49	−6.25	−6.87	−4.81	−3.52	−1.56

Notes: Lower PRIAL = better forecast accuracy. No PCs identifies the corresponding MinT or IComb benchmark.

Findings:

- Component augmentation through FLAP consistently has better accuracy than MinT_{shr}
- FLAIC achieves better absolute accuracy than FLAP in the reported matched comparisons.
- The benefits of component augmentation depend on the reconciliation method.

** More details in appendix slides*

2.4.9 *Robustness: Using scaled errors*

- Series at hierarchy levels differ substantially in scale.
- Compute mean squared *scaled* forecast error (MSSFE):

$$\text{MSSFE}_{m,h} = \frac{1}{N_h n} \sum_{\tau \in \mathcal{O}_h} \sum_{i=1}^{n} \frac{\left(\tilde{e}_{\tau,h,i}^{(m)}\right)^2}{d_{\tau,i}}$$

using two scaling denominators from 120-month training window:

1. Seasonal naïve forecast: $d_{\tau,i}^{(SN)} = \frac{1}{108} \sum_{t=\tau-107}^{\tau} \left(y_{t,i} - y_{t-12,i}\right)^2$
2. Variance standardized: $d_{\tau,i}^{(V)} = \frac{1}{120} \sum_{t=\tau-119}^{\tau} \left(y_{t,i} - \bar{y}_{\tau,i}\right)^2$

Finding	**Raw error measure**	**Both scaled-errors measures**
Low/moderate augmentation can help	Yes	Yes (smaller than raw squared errors)
High augmentation generally hurts	Yes	Yes
Best-performing FLAIC variant	Lasso	Ridge

Concluding remarks: contribution & significance

- Research question: *To what extent* can forecasts based on constructed linear components improve the accuracy of hierarchical forecasts when incorporated into the Information Combination framework?

- Answer: It can, under selected specifications — but improvements in forecast accuracy are not consistently statistically significant.

- Our contribution is mainly methodological:
  - Propose FLAIC: incorporates separately forecast principal components into IComb framework, while maintaining coherence and without requiring additional underlying data.
  - Develop a hierarchy-adapted FLAP benchmark, incorporating both component and aggregation constraints.

- Implications – established via empirical investigation
  - Low/moderate augmentation can improve accuracy of FLAIC relative to IComb, particularly under ridge; very high augmentation generally deteriorates FLAIC's relative forecast performance.
  - Benefits of component augmentation depend on PCA specification, regularisation, and reconciliation method.

Limitations and future research

Limitations of our research:

- One empirical dataset (Australian domestic tourism) and ETS forecasts
- Unclear when component forecasts offer genuinely useful complementary information
- Predefined PCA specification, component augmentation dimension and one IComb implementation.

Identifying the conditions under which component augmentation provides a reliable improvement in forecast accuracy relative to IComb is an important direction for future research:

- Monte Carlo simulations varying hierarchy structure, base forecasting models, common signals, forecast error correlation and dimensionality
- Use different IComb framework variants (example, without standardisation)
- If best FLAIC variant is an empirical question, then may need data-driven selection for PCA, augmentation dimension and regularisation method

Appendix

2.0 Empirical Investigation

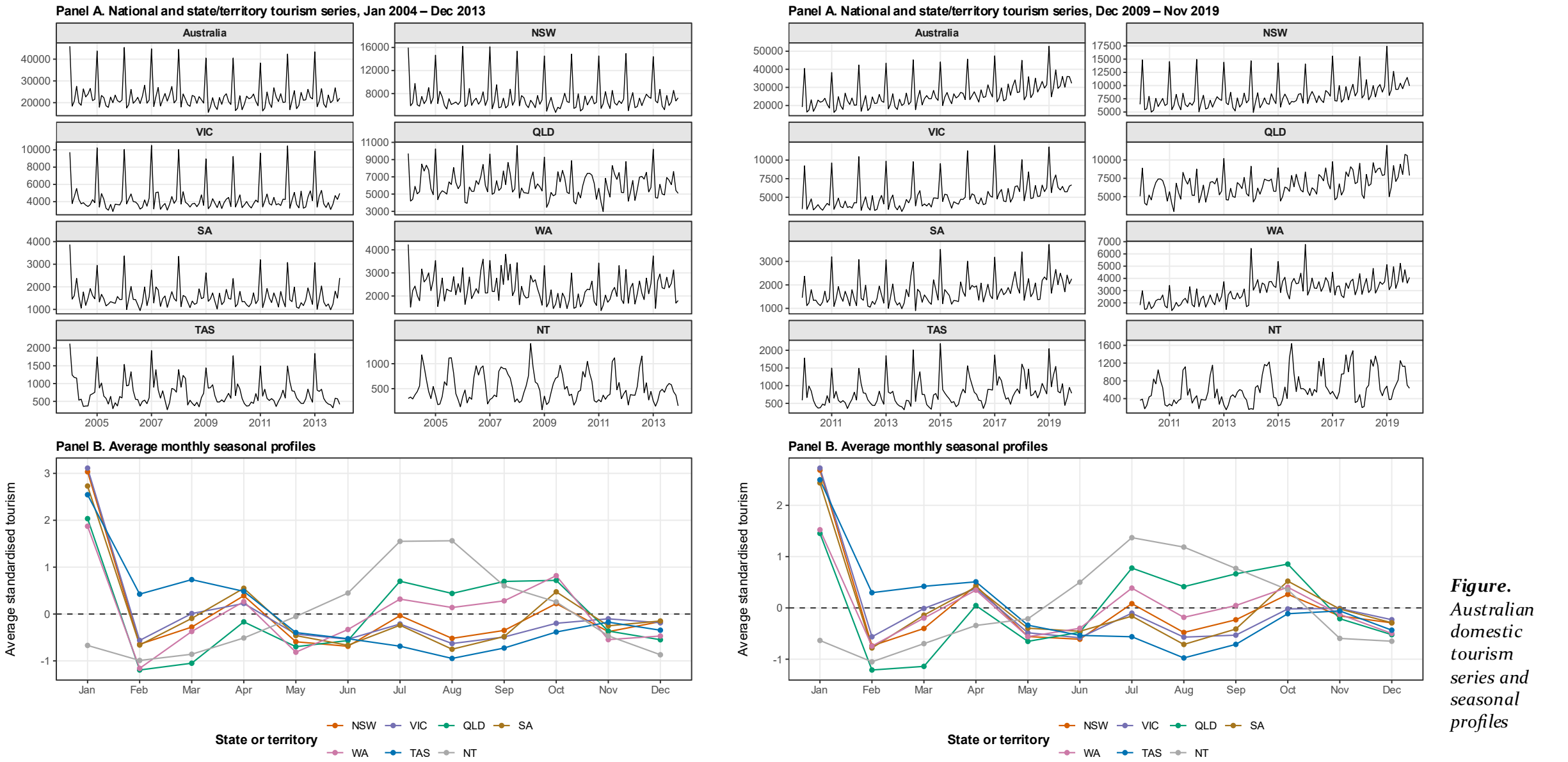
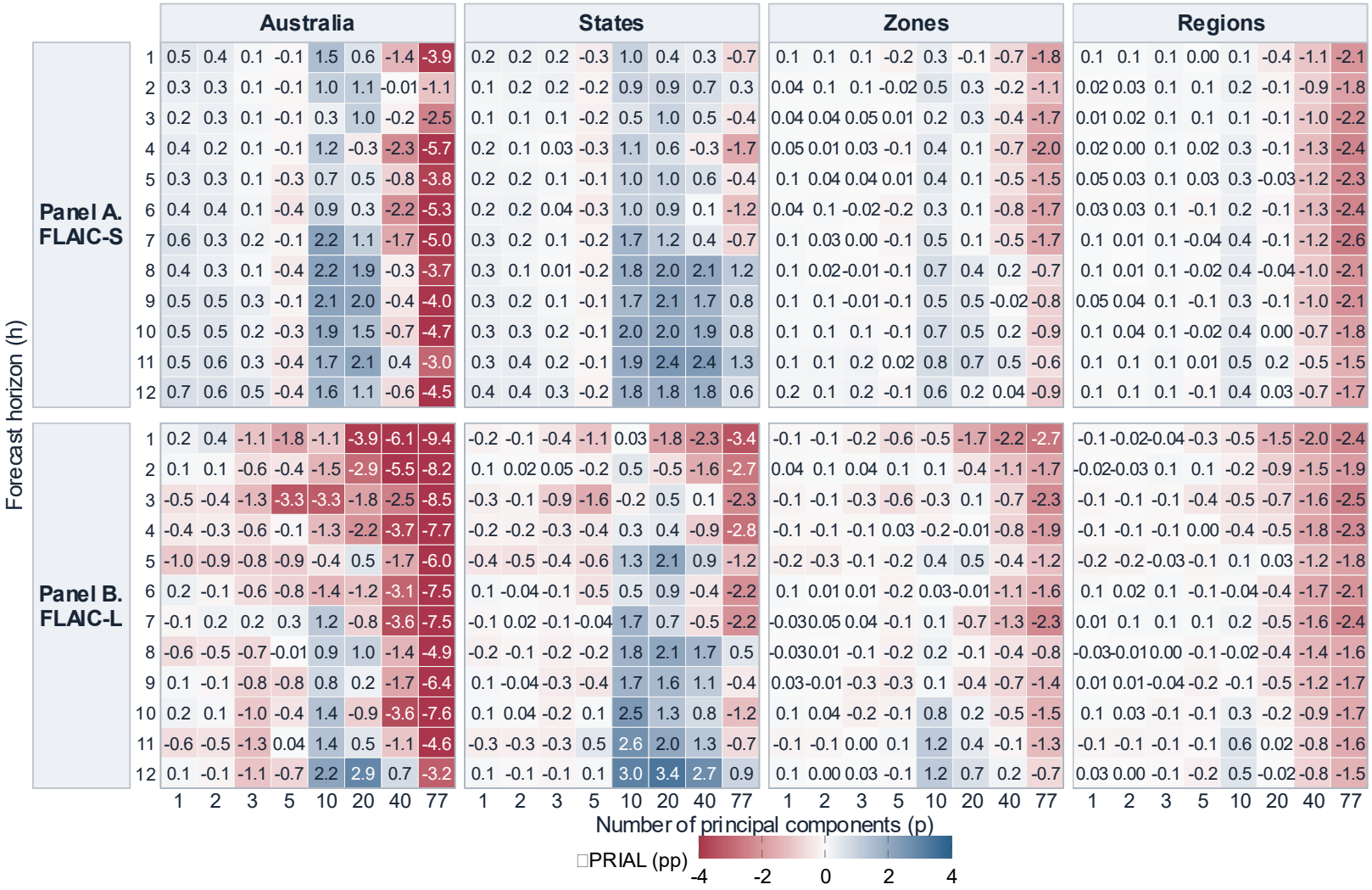


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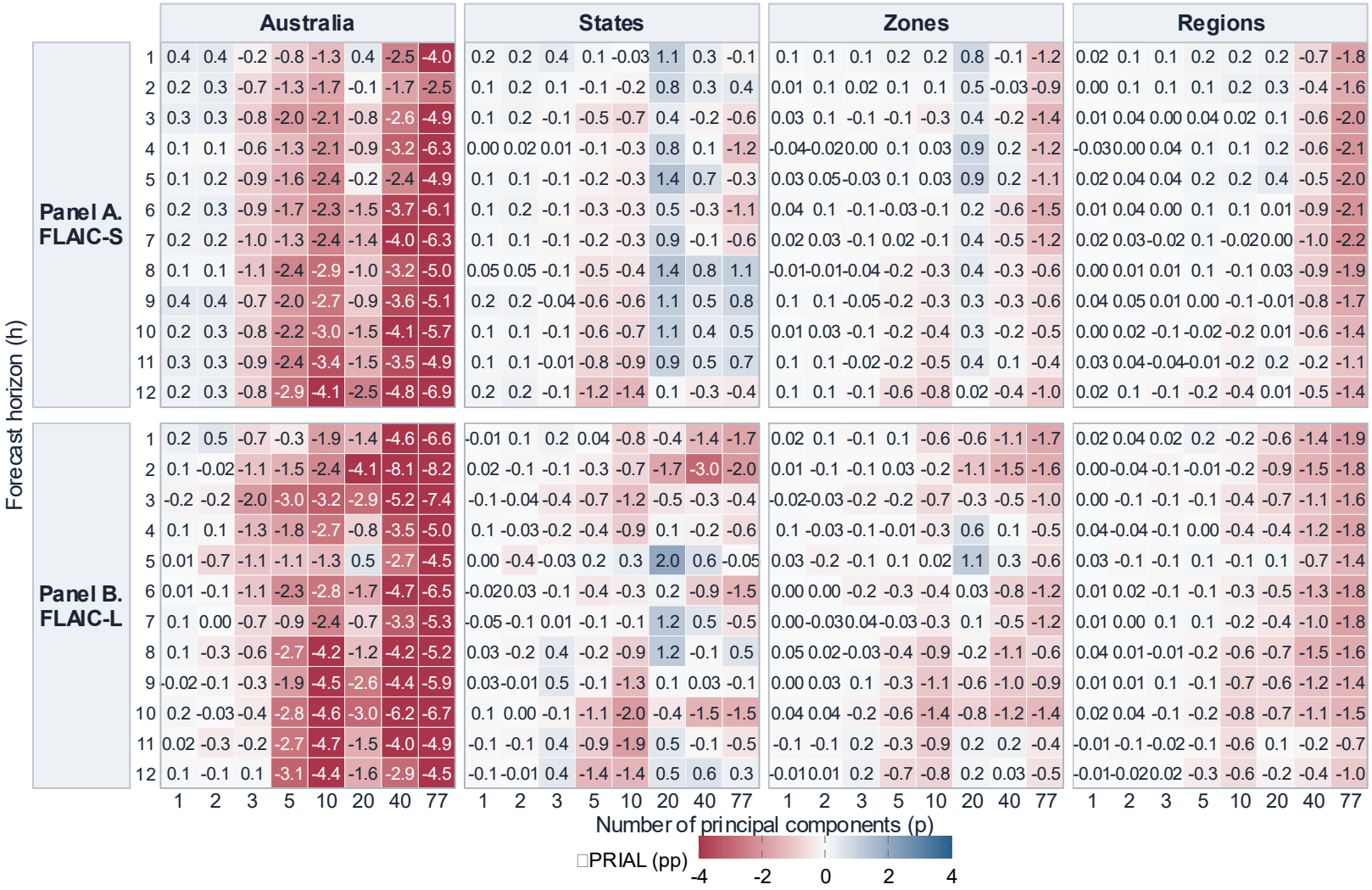
2.4.4 Δ PRIAL and PRIAL by hierarchy level



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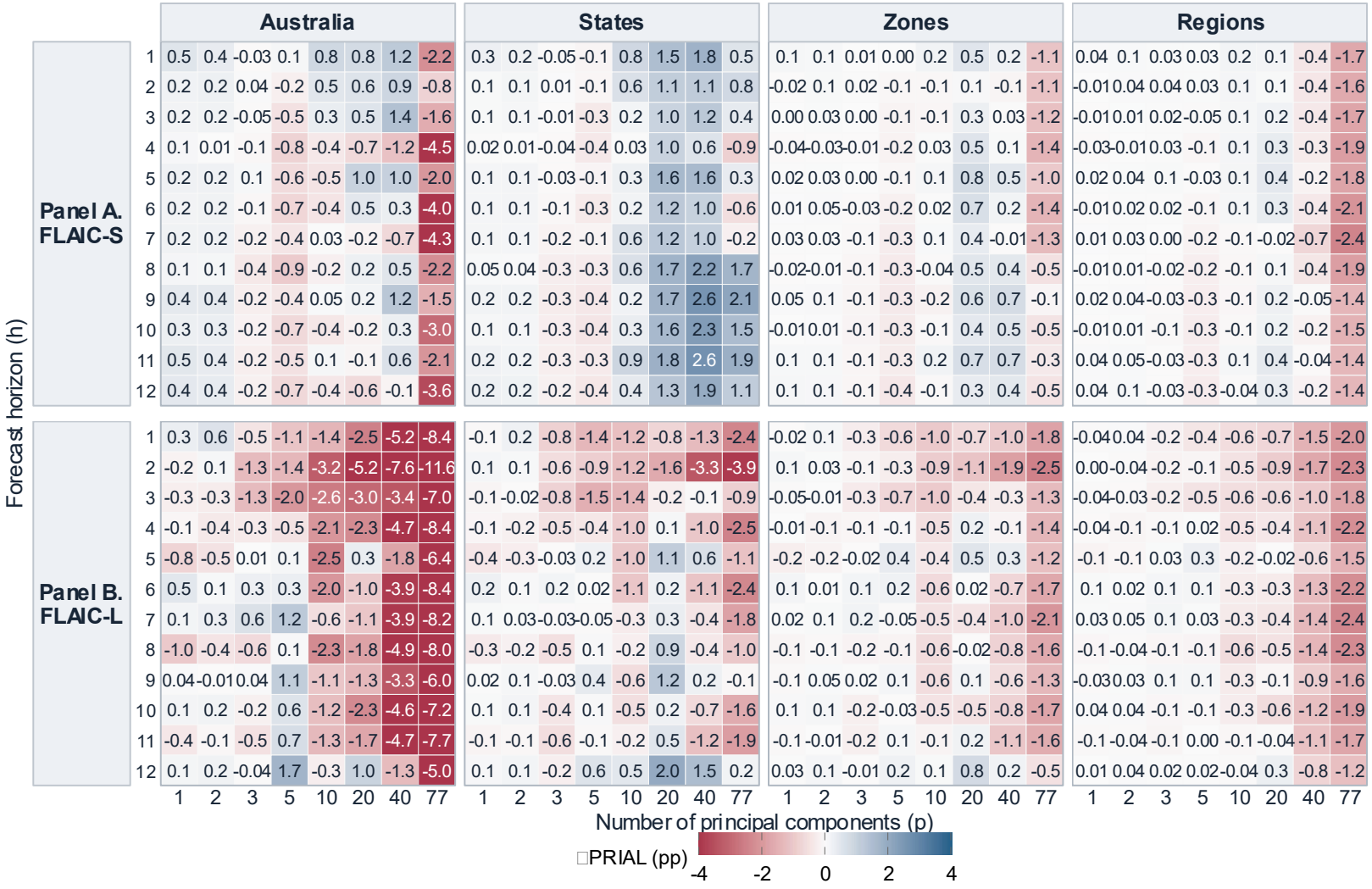
2.4.4 Δ PRIAL and PRIAL by hierarchy level



- Heterogeneous effects across hierarchy levels
  - ✓ States tend to benefit more
- Higher dimensional component augmentation mostly deteriorates relative forecast performance
- Relative to base forecast (PRIAL):
  - Best PCA specification, p, ridge/lasso vary by level

Figure: *ΔPRIAL*: FLAIC relative to matched IComb (**using uncentred, unscaled PCA**), computed for each hierarchy level. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L, and positive values favour FLAIC over matched IComb.

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2.4.5 Statistical significance visualised via MCB

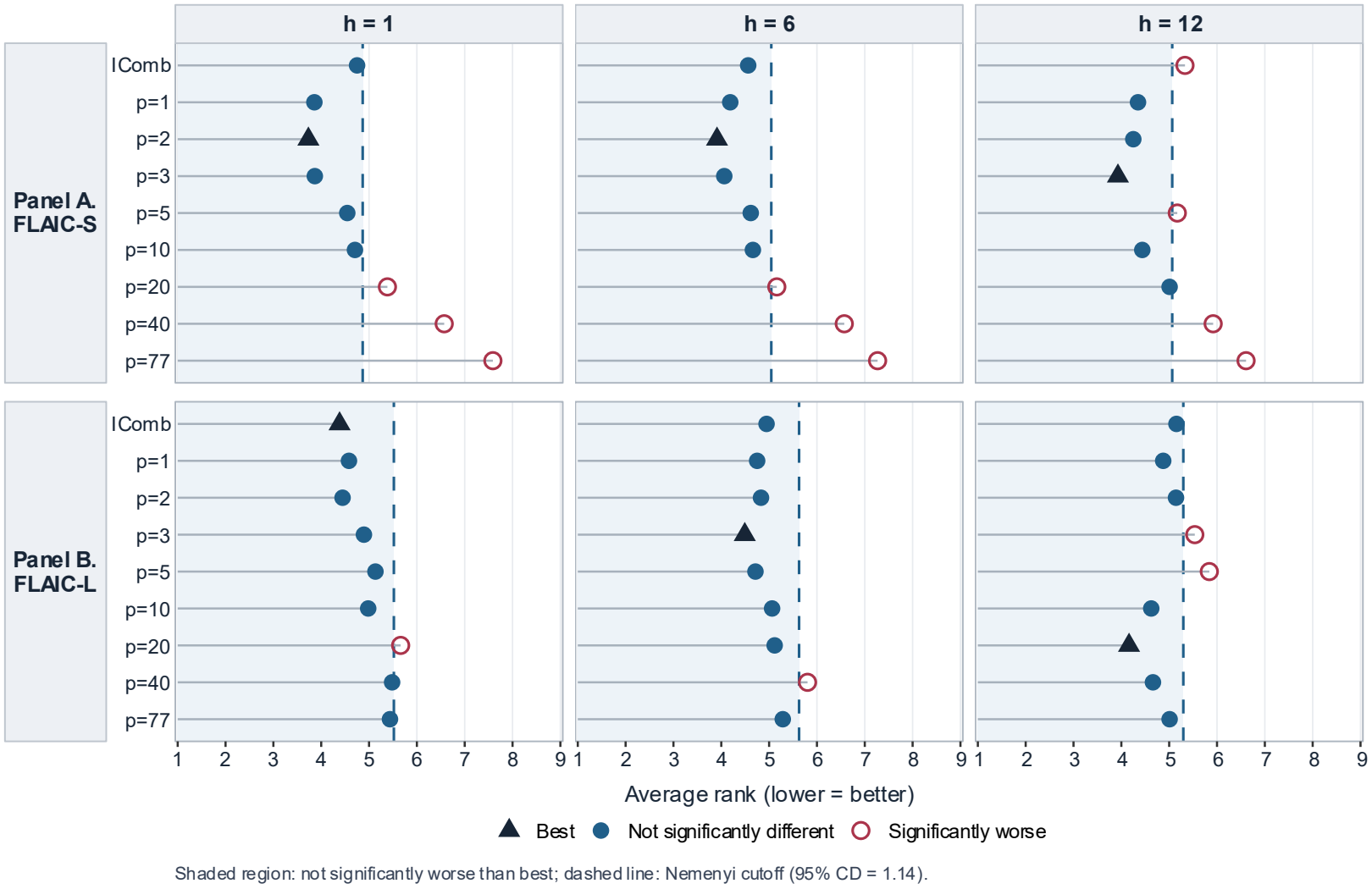


Figure: Multiple comparison with best (MCB): comparison of average ranks against best-rank method. Uses average MSFE ranks across 112 hierarchy series at $h = 1,6,12$ for **centred, scaled PCA specification**.

$$\text{MSFE}_{m,h,i} = \frac{1}{N_h} \sum_{\tau \in \mathcal{O}_h} \left(\tilde{e}_{\tau,h,i}^{(m)} \right)^2$$

- Rank each series across methods to get a distribution of ranks for each method. Each method is summarised by its average rank across series.
  (Friedman 1937, 1939; Nemenyi 1963; Hollander et al. 2013; Koning et al. 2005)

Considering other 3 PCA specs too:

- Best-ranked method tends to be low augmentation
- Limited evidence that improvement in forecast accuracy under FLAIC is statistically significant (especially lasso)
- High-dimensional component augmentation typically statistically significantly worse (especially ridge)

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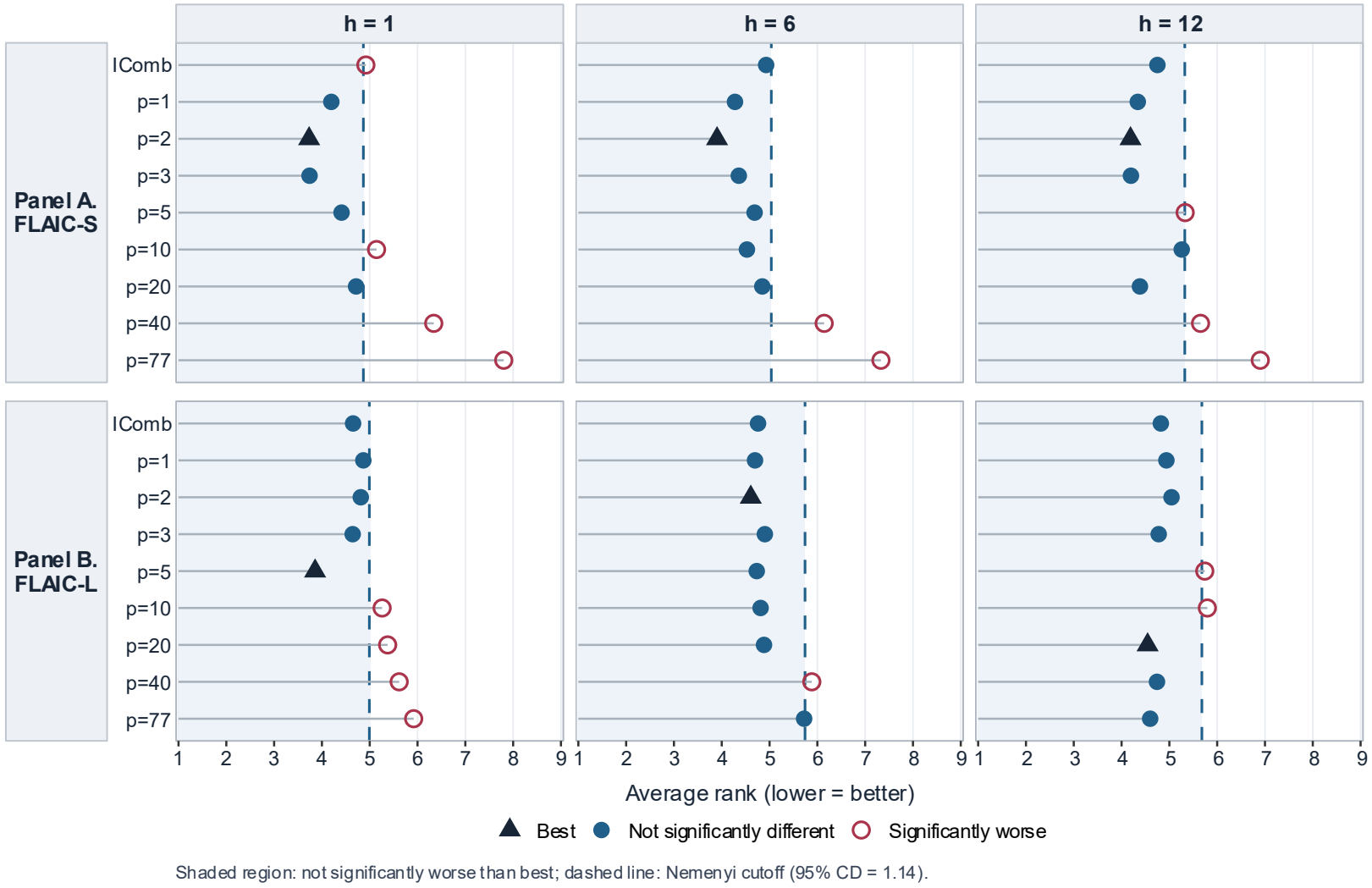


Figure: Multiple comparison with best (MCB): comparison of average ranks against best-rank method. Uses average MSFE ranks across 112 hierarchy series at $h = 1,6,12$ for **uncentred, unscaled PCA specification**.

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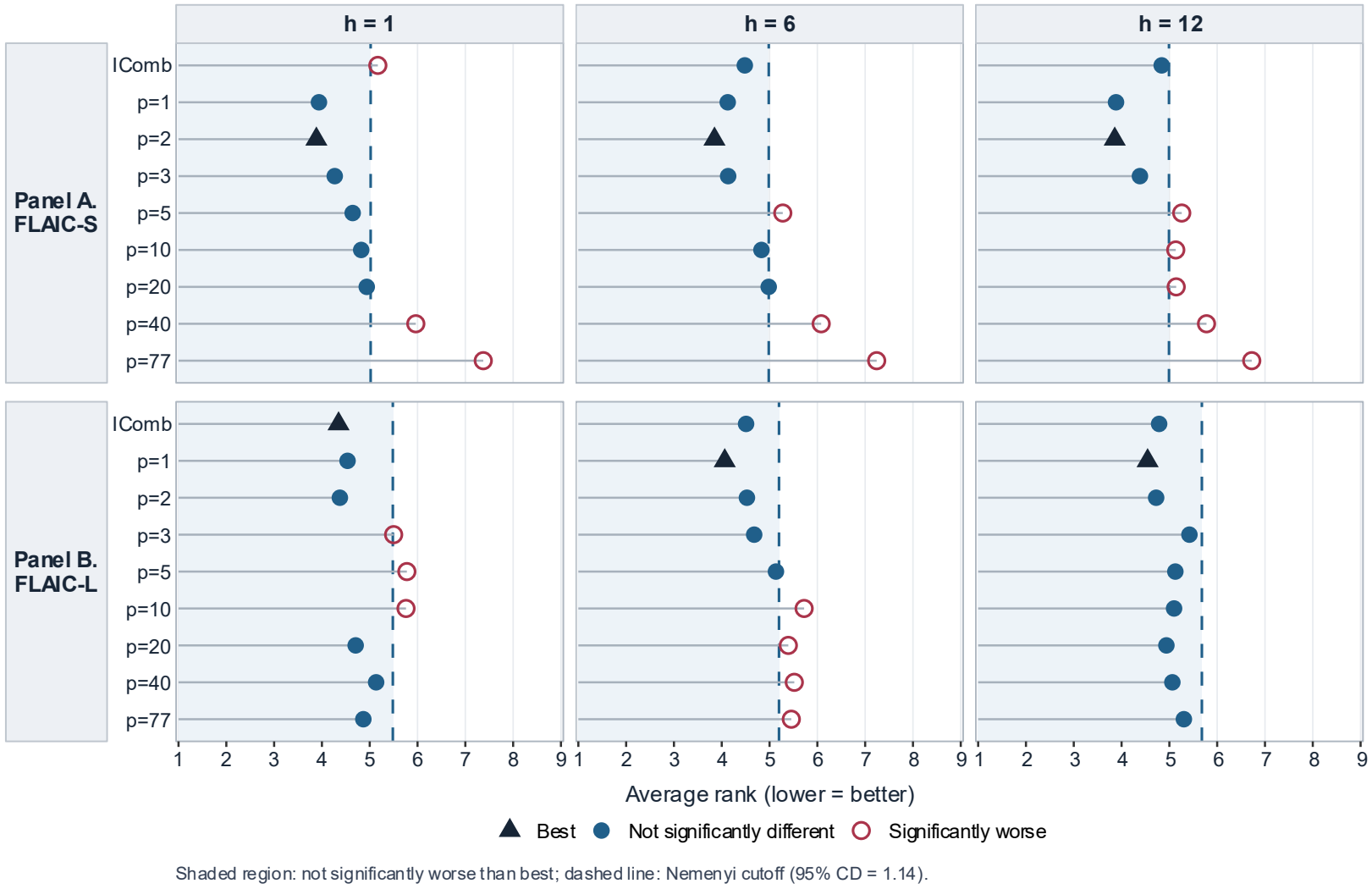


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2.4.7 *Hierarchy-adapted FLAP benchmark*

Same base + component forecasts → Hierarchy adapted FLAP benchmark: Covariance-based oblique projection → Coherent forecasts

- Enforces hierarchy aggregation constraints and uses same component relationships.
- Reduces to MinT when $p = 0$.

(1) Write our component augmented vector as:

$$\begin{bmatrix} \mathbf{c_t}^{(\delta_C,\delta_S)} + \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{\delta_C}^{-\delta_S} \delta_C \mathbf{S}\bar{\mathbf{b}} \\ \mathbf{y_t} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{(\delta_C)}^{-\delta_S} \mathbf{S} \\ \mathbf{S} \end{bmatrix} \mathbf{b_t}$$

(2) We therefore have

$$\mathbf{z}_t^* = \begin{bmatrix} \mathbf{v}_t^{(\delta_C,\delta_S)} \\ \mathbf{a}_t \\ \mathbf{b}_t \end{bmatrix} = \begin{bmatrix} \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{(\delta_C)}^{-\delta_S} \mathbf{S} \\ \mathbf{A} \\ \mathbf{I}_{n_b} \end{bmatrix} \mathbf{b}_t$$

where $\mathbf{S}_p^* = \begin{bmatrix} \boldsymbol{\Phi}_p^* \\ \mathbf{I}_{n_b} \end{bmatrix}$ and $\boldsymbol{\Phi}_p^* = \begin{bmatrix} \boldsymbol{\Phi}_p^{(\delta_C,\delta_S)} \mathbf{D}_{(\delta_C)}^{-\delta_S} \mathbf{S} \\ \mathbf{A} \end{bmatrix} \in \mathbb{R}^{(p+n_a)\times n_b}$

(3) Hence, we use

$$\mathbf{G}_{h,p}^* = \left[(\mathbf{S}_p^*)'(\mathbf{W}_h^*)^{-1}\mathbf{S}_p^*\right]^{-1}(\mathbf{S}_p^*)'(\mathbf{W}_h^*)^{-1}$$

to obtain

$$\tilde{\mathbf{y}}_{t+h|t}^{\text{FLAP}} = \mathbf{S}\tilde{\mathbf{b}}_{t+h|t}^{\text{FLAP}} = \mathbf{S}\mathbf{G}_{h,p}^* \hat{\mathbf{z}}_{t+h|t}^*$$

2.4.8 Robustness: Using scaled errors

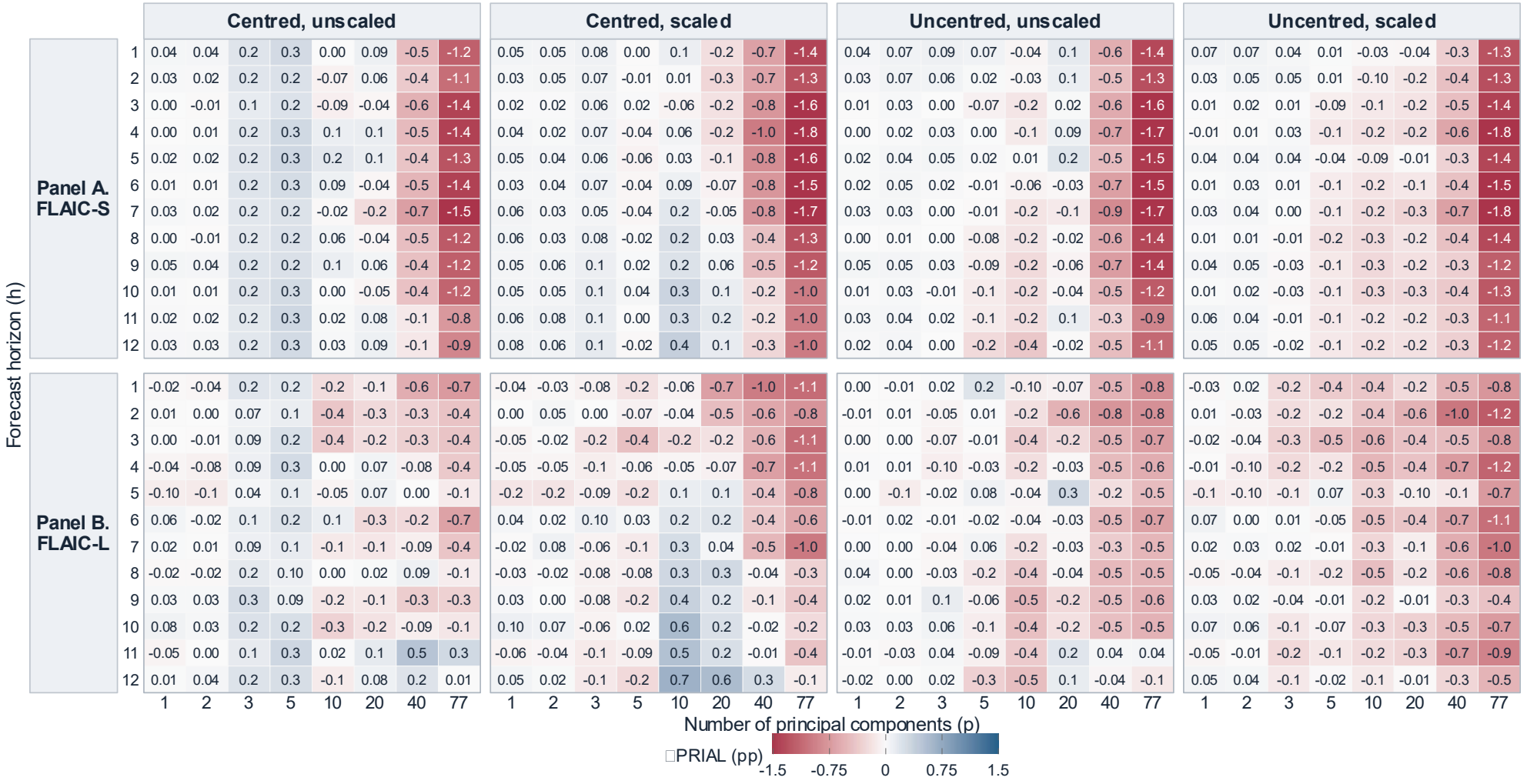


Figure: PRIAL of FLAIC relative to IComb: Average across 112 hierarchy series. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L. Uses **seasonal-naïve scaled errors** to compute MSSFE, the PRIAL

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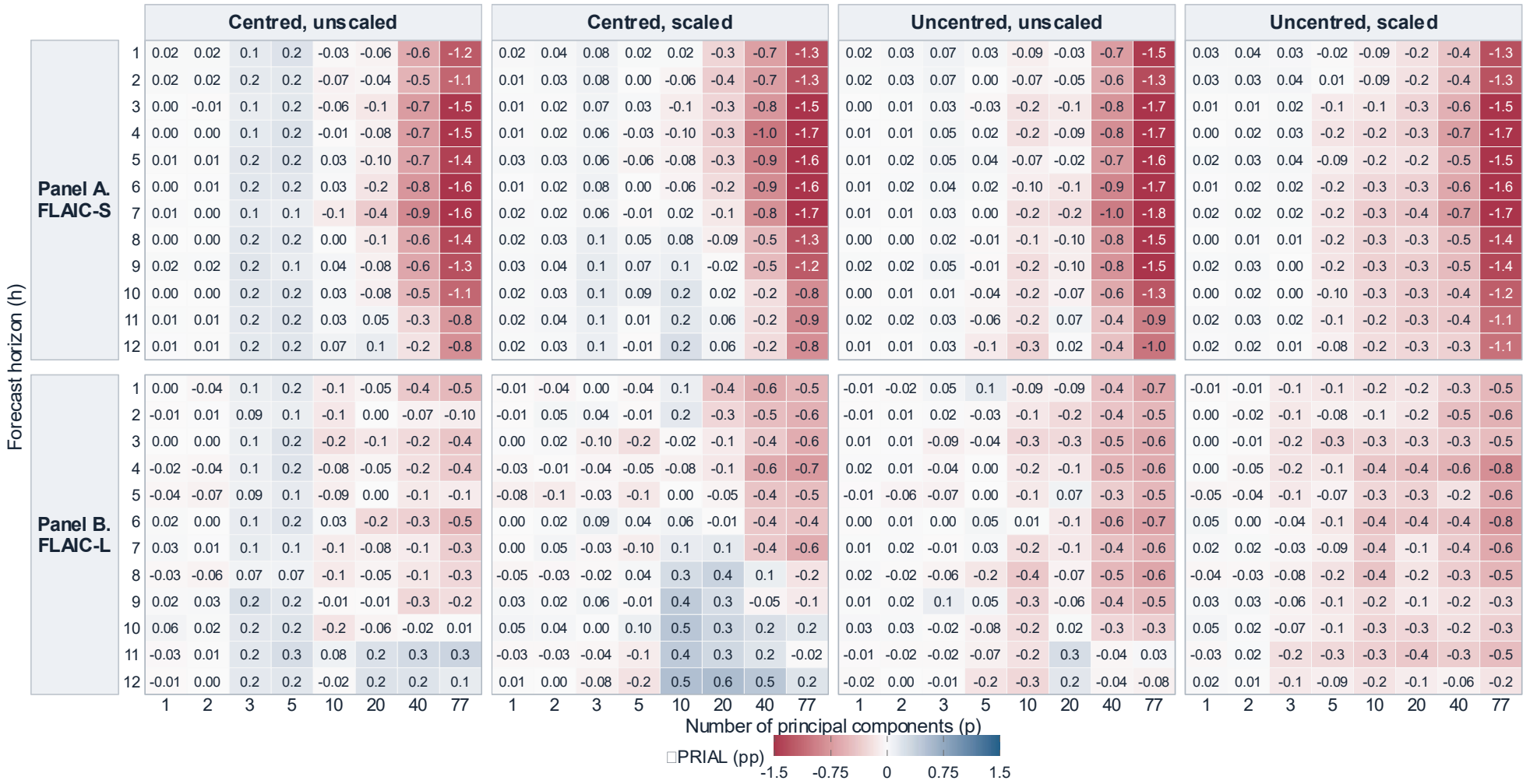


Figure: PRIAL of FLAIC relative to IComb: Average across 112 hierarchy series. FLAIC-S and FLAIC-L are compared against matching IComb-S and IComb-L. Uses **variance standardised scaled errors** to compute MSSFE, the PRIAL

MinT details

- Key benchmark: Minimum Trace (MinT) reconciliation (Wickramasuriya et al., 2019).

  Under unbiasedness of base forecasts:

$$\mathbf{G}_h^{\text{MinT}} = \left(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{W}_h^{-1}$$

  where $\mathbf{W}_h$ is base forecast error covariance matrix

  - Unbiasedness constraint (which is constraint needed to preserve unbiasedness of reconciled forecasts if base forecasts are unbiased): $\mathbf{G}_h\mathbf{S} = \mathbf{I}_{n_b}$
  - Minimise trace of $\text{var}\left[\tilde{\mathbf{e}}_{t+h|t}\right]$ $\quad (\tilde{\mathbf{e}}_{t+h|t} = \mathbf{y}_{t+h} - \tilde{\mathbf{y}}_{t+h|t})$
  - $\mathbf{G}_h^{\text{MinT}} = \left(\mathbf{S}'\mathbf{W}_h^{-1}\mathbf{S}\right)^{-1}\mathbf{S}'\mathbf{W}_h^{-1}$, $\quad \mathbf{W}_h = \text{var}\left[\hat{\mathbf{e}}_{t+h|t}\right]$ $\quad (\hat{\mathbf{e}}_{t+h|t} = \mathbf{y}_{t+h} - \hat{\mathbf{y}}_{t+h|t})$
  - $\mathbf{W}_h$ estimated from in-sample forecast errors with $\mathbf{W}_h = \text{k}_\text{h}\widehat{\mathbf{W}}_1$

MinT details (cont.)

- MinT has OLS and WLS as special cases:
  - $\mathbf{W}_h$ is base forecast error covariance matrix
  - If we assume $\mathbf{W}_h = \mathrm{k_h}\mathbf{I}_n$, then we have OLS case
  - If we assume $\mathbf{W}_h = \mathrm{k_h}\,\mathrm{diag}(\widehat{\mathbf{W}}_1)$, then we have WLS (variance scaling) case (Hyndman et al., 2016)
  - If we assume $\mathbf{W}_h = \mathrm{k_h}\,\mathrm{diag}(\mathbf{S1})$, then we have WLS (structural scaling) case (Athanasopoulos et al., 2017)
  - MinT gives same answer as GLS even though different loss function being minimised (Wickramasuriya, 2021)
  - MinT and its special cases are considered projections as $\mathbf{SG}_h$ is a *projection* onto $\mathrm{span}(\mathbf{S})$. In these cases, $\mathbf{SG}_h$ is a projection matrix as it meets idempotence property: $(\mathbf{SG}_h)^2 = \mathbf{SG}_h$